

AirFret Sound, Note & Chord Calculations

UCLA Engineering 1 TR HAcK 2026 - Technical Calculation Sheet

Purpose. This document records the numerical calculations behind the AirFret note engine, chord bank, strum samples, and sound-effect banks. Values are based on the current AirFret Python code and the guitar bank builder.

Parameter	AirFret value	Meaning
Sample rate	22,050 Hz	22,050 samples are produced each second.
PCM depth	16-bit signed	2 bytes per mono sample.
I2S output	Stereo	The Pico duplicates mono samples to Left and Right.
Reference pitch	A4 = 440 Hz	Equal-tempered pitch reference.
Chord sample duration	1.400 s	Each pre-rendered D1/D2/U1 chord file.
Note-mode octave	C4 through C5	8 keypad notes = one complete octave.

Core pitch formula

$$f(m) = 440 \times 2^{((m - 69) / 12)}$$

where **m** is the MIDI note number. MIDI 69 is A4, so $f(69) = 440$ Hz. Increasing MIDI by 12 doubles frequency; decreasing it by 12 halves frequency.

PCM / data-rate calculations

Mono PCM data rate = 22,050 samples/s x 2 bytes/sample = **44,100 bytes/s**. Stereo I2S data rate = 22,050 x 2 bytes x 2 channels = **88,200 bytes/s**.

1. Note-Mode Pitch Calculations

AirFret maps keypad keys 1-8 to MIDI 60, 62, 64, 65, 67, 69, 71, 72. CLEAN uses the base MIDI value; OCTAVE adds 12 semitones; SYNTH uses the same base pitch with a triangle waveform.

Key	Note	MIDI	Frequency (Hz)	Clean loop samples	Clean buffer	Octave note	Octave Hz	Octave samples	Octave buffer
1	C4	60	261.63	337	1348 B	C5	523.25	169	676 B
2	D4	62	293.66	300	1200 B	D5	587.33	150	600 B
3	E4	64	329.63	268	1072 B	E5	659.26	134	536 B
4	F4	65	349.23	253	1012 B	F5	698.46	126	504 B
5	G4	67	392.00	225	900 B	G5	783.99	113	452 B
6	A4	69	440.00	200	800 B	A5	880.00	100	400 B
7	B4	71	493.88	179	716 B	B5	987.77	89	356 B
8	C5	72	523.25	169	676 B	C6	1046.50	84	336 B

Loop-size calculation used by CLEAN / OCTAVE / SYNTH

$\text{samples} = \text{round}(4 \times 22,050 / \text{frequency})$ $\text{buffer bytes} = \text{samples} \times 4 \text{ bytes}$

The factor of 4 means each loop contains four waveform cycles. Each runtime note buffer stores a 16-bit Left sample plus a matching 16-bit Right sample, so one stereo frame is 4 bytes.

CLEAN waveform: $\text{sample}[i] = 4000 \times \sin(2\pi \times 4 \times i / \text{samples})$.

OCTAVE waveform: same sine-wave calculation after $\text{MIDI} := \text{MIDI} + 12$, therefore frequency doubles.

SYNTH waveform: same base frequency but a triangle waveform with amplitude 3600, producing a brighter electronic timbre.

2. Pitched WAH and LASER Note Effects

WAH and LASER are pre-rendered note-specific RAW files. The Pico does not perform real-time pitch shifting; it selects the file that already corresponds to the requested note. This protects audio timing while the instrument also handles controls and strumming.

Key	Target note	Target Hz	Semitones from A4	Frequency ratio vs A4	WAH file	LASER file
1	C4	261.63	-9	0.5946	WAH_C4.raw	LASER_C4.raw
2	D4	293.66	-7	0.6674	WAH_D4.raw	LASER_D4.raw
3	E4	329.63	-5	0.7492	WAH_E4.raw	LASER_E4.raw
4	F4	349.23	-4	0.7937	WAH_F4.raw	LASER_F4.raw
5	G4	392.00	-2	0.8909	WAH_G4.raw	LASER_G4.raw
6	A4	440.00	+0	1.0000	WAH_A4.raw	LASER_A4.raw
7	B4	493.88	+2	1.1225	WAH_B4.raw	LASER_B4.raw
8	C5	523.25	+3	1.1892	WAH_C5.raw	LASER_C5.raw

Pitch-ratio formula

$\text{ratio} = 2^{(\text{semitones} / 12)}$ example for C4: semitones = 60 - 69 = -9; $\text{ratio} = 2^{(-9/12)} = 0.5946$; target frequency = $440 \times 0.5946 = 261.63$ Hz

Each WAH note file is 27,342 bytes, so duration = $27,342 / 44,100 = 0.620$ s. Eight WAH notes use 218,736 bytes.

Each LASER note file is 75,984 bytes, so duration = $75,984 / 44,100 = 1.723$ s. Eight LASER notes use 607,872 bytes.

3. Guitar Chord Voicings and Frequencies

The chord bank models six guitar strings in the order Low E, A, D, G, B, High E. "x" means that string is muted. MIDI values are converted to frequency with the same A4=440 Hz formula.

Chord	Low E	A	D	G	B	High E	Sounding frequencies (Hz)
C major	x	C3 (48)	E3 (52)	G3 (55)	C4 (60)	E4 (64)	C3 130.8, E3 164.8, G3 196.0, C4 261.6, E4 329.6
D minor	x	x	D3 (50)	A3 (57)	D4 (62)	F4 (65)	D3 146.8, A3 220.0, D4 293.7, F4 349.2
E minor	E2 (40)	B2 (47)	E3 (52)	G3 (55)	B3 (59)	E4 (64)	E2 82.4, B2 123.5, E3 164.8, G3 196.0, B3 246.9, E4 329.6
F major	F2 (41)	C3 (48)	F3 (53)	A3 (57)	C4 (60)	F4 (65)	F2 87.3, C3 130.8, F3 174.6, A3 220.0, C4 261.6, F4 349.2
G major	G2 (43)	B2 (47)	D3 (50)	G3 (55)	B3 (59)	G4 (67)	G2 98.0, B2 123.5, D3 146.8, G3 196.0, B3 246.9, G4 392.0
A minor	x	A2 (45)	E3 (52)	A3 (57)	C4 (60)	E4 (64)	A2 110.0, E3 164.8, A3 220.0, C4 261.6, E4 329.6
B diminished	x	B2 (47)	F3 (53)	B3 (59)	D4 (62)	F4 (65)	B2 123.5, F3 174.6, B3 246.9, D4 293.7, F4 349.2
B minor	x	B2 (47)	F#3 (54)	B3 (59)	D4 (62)	F#4 (66)	B2 123.5, F#3 185.0, B3 246.9, D4 293.7, F#4 370.0
D major	x	x	D3 (50)	A3 (57)	D4 (62)	F#4 (66)	D3 146.8, A3 220.0, D4 293.7, F#4 370.0
F# diminished	F#2 (42)	C3 (48)	F#3 (54)	A3 (57)	C4 (60)	F#4 (66)	F#2 92.5, C3 130.8, F#3 185.0, A3 220.0, C4 261.6, F#4 370.0
E diminished	E2 (40)	A#2 (46)	E3 (52)	G3 (55)	A#3 (58)	E4 (64)	E2 82.4, A#2 116.5, E3 164.8, G3 196.0, A#3 233.1, E4 329.6
G minor	G2 (43)	D3 (50)	G3 (55)	A#3 (58)	D4 (62)	G4 (67)	G2 98.0, D3 146.8, G3 196.0, A#3 233.1, D4 293.7, G4 392.0
Bb major	x	A#2 (46)	F3 (53)	A#3 (58)	D4 (62)	F4 (65)	A#2 116.5, F3 174.6, A#3 233.1, D4 293.7, F4 349.2

4. Chord Sound-Bank Calculations

AirFret pre-renders three performances for each chord: two down-strums (D1, D2) and one up-strum (U1). The generator uses a Karplus-Strong-style physical-string model plus staggered string starts and body-resonance delays.

Calculation	Result
Samples per 1.4 s chord file	$22,050 \times 1.4 = 30,869$ samples
Bytes per mono chord file	$30,869 \times 2 = 61,738$ bytes
Files per chord	$D1 + D2 + U1 = 3$ files
Bytes per chord	$61,738 \times 3 = 185,214$ bytes
Number of chord voicings	13
Nominal complete chord bank	$185,214 \times 13 = 2,407,782$ bytes (~2.30 MiB)

Karplus-Strong string period

```
period ~= int(22,050 / frequency - 0.5)
```

Example string	Frequency	Calculated delay period
E2	82.41 Hz	267 samples
C3	130.81 Hz	168 samples
G3	196.00 Hz	112 samples
C4	261.63 Hz	83 samples
E4	329.63 Hz	66 samples
G4	392.00 Hz	55 samples

Down strum timing: strings are started from low to high, with approximately 5-10 ms added between adjacent strings. **Up strum:** high to low, with about 4-8 ms between strings.

Performance strength: down-strums use a randomized 96-108% strength range; up-strums use 82-98%. This is one reason D1/D2/U1 do not sound identical.

Body resonance: the builder mixes delayed reflections of 71, 149, and 233 samples at 10%, 6%, and 4%, plus a low-pass body term at 7%. A 100 ms fade is applied at the end of each file.

5. Scale and Song Chord Maps

These are the chord sequences stored in the AirFret chord-mode mapping.

Mode / song	Chord sequence
C major	C major -> D minor -> E minor -> F major -> G major -> A minor -> B diminished
G major	G major -> A minor -> B minor -> C major -> D major -> E minor -> F# diminished
A minor	A minor -> B diminished -> C major -> D minor -> E minor -> F major -> G major
D minor	D minor -> E diminished -> F major -> G minor -> A minor -> Bb major -> C major
Country Roads	G major -> E minor -> D major -> C major -> G major
Never A	G major -> C major -> D major -> G major -> C major -> D major -> G major
Never B	C major -> G major -> D major -> G major -> C major -> D major -> G major
Stand By Me	G major -> E minor -> C major -> D major

Stand By Me AirFret mode

G major -> E minor -> C major -> D major -> repeat. The joystick advances or selects chords depending on the current performance control logic, while gyro DOWN/UP selects the corresponding D1/D2 or U1 strum performance.

6. Standalone Performance Effects

WAH, PLUCK, and WHIP are stored as fixed pre-rendered mono signed-16-bit RAW samples at 22,050 Hz. Their duration is calculated directly from file size because mono 16-bit audio uses 44,100 bytes each second.

Effect	RAW size	Duration calculation	Approx. duration
WAH	85,490 B	$85,490 / 44,100$	1.939 s
PLUCK	11,732 B	$11,732 / 44,100$	0.266 s
WHIP	87,320 B	$87,320 / 44,100$	1.980 s

Direct-stream playback

For clean playback, AirFret reads the 16-bit mono RAW sample and duplicates the same two bytes into the Left and Right I2S channels. This avoids expensive live filtering or pitch shifting on the Pico.

mono sample [low byte, high byte] -> stereo frame [low, high, low, high]

7. Calculation Summary for Design Review

Feature	Calculation / evidence
One octave	C4 (261.63 Hz) through C5 (523.25 Hz), 8 keypad notes.
Equal temperament	$f = 440 \times 2^{((\text{MIDI}-69)/12)}$.
Clean note	Sine-wave buffer, amplitude 4000, 4 cycles per loop.
Octave note	MIDI + 12 -> exactly 2x base frequency.
Synth note	Triangle-wave buffer at the same requested fundamental frequency.
Pitched WAH	8 pre-rendered WAH samples corresponding to C4-D4-E4-F4-G4-A4-B4-C5.
Pitched LASER	8 pre-rendered LASER samples using equal-tempered semitone pitch ratios.
Chord bank	13 guitar voicings x 3 strum performances = 39 RAW chord files.
Chord sample size	61,740 bytes per 1.4 s mono 16-bit file at 22,050 Hz.
Performance FX	WAH, PLUCK, WHIP are separate pre-rendered sound samples.
I2S stream	Mono source is duplicated to stereo: 88,200 output bytes/s.

Source code basis: AirFret main program (note/effect mappings and note synthesis) and `build_bank.py` (guitar voicings and strum synthesis).

Note: This is an engineering calculation/documentation sheet for the AirFret prototype; it describes the programmed audio system rather than an acoustic measurement of the physical speaker/amplifier.